RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Design

GE23627 Design Thinking and Innovation

Ideate Phase – Ideation Workshop Sheet

Team Information

Team Name:	**Student 1(Roll No and Name)**	**Student 2(Roll No and Name)**
Team Guardians	241701025 MAAHESH S	241701042 PRADEEP KUMAR R

Problem Statement:

Hostel students face difficulties with manual gate-pass applications, delayed approvals, unreliable parent verification, manual entry/exit records, late-return monitoring, and limited emergency support. Parents may also not receive timely information about their child's hostel movement and safety.

Target Users:

Primary Users:

- Hostel Students
- Parents / Guardians

Secondary Users:

- Wardens
- Security Staff
- Hostel Administration
- College Management

Brainstorming Rules

1. Focus on generating many ideas.
2. Do not criticize any idea during brainstorming.
3. Encourage creative and unusual ideas.
4. Build on ideas proposed by teammates.
5. Write or sketch ideas

Idea Generation (Brainstorming)

Idea No	Student 1 Idea	Student 2 Idea
1	**Digital Gate Pass:** Students can apply for gate passes online and track approval status without using paper forms.	**Smart Gate Pass:** A mobile system allows students to submit outing details, reducing paperwork and making the approval process faster.
2	**Parent Approval:** Parents receive gate-pass requests and approve or reject them through the app or automated call.	**Parent Notification:** Parents receive instant notifications about gate-pass requests, approvals, exits, entries, and late returns.
3	**QR Gate Verification:** Approved students receive a temporary QR code for quick and secure hostel exit and entry verification.	**Secure QR Pass:** The system generates an encrypted temporary QR code that security staff scan to record student movement.
4	**Safety Monitoring:** Parents receive student movement updates, late-return alerts, and safety notifications during approved outings.	**Emergency SOS:** Students can use an SOS button to alert parents and authorized emergency contacts when facing safety problems.
5	**Biometric Verification:** Student identity is verified using fingerprint or device biometrics before generating the secure QR code.	**Location Awareness:** Parents can view the student's location during an approved outing to improve safety and provide timely assistance.

Idea Voting (Dot Voting)

Idea	Student 1 Vote	Student 2 Vote	Total Votes
Digital Gate Pass	✓	✓	2
Parent Approval	✓	✓	2
QR Verification	✓	✓	2
Safety & Tracking	✓		1
Biometric Verification	✓	✓	2

Top Idea Selected

Selected Idea:

Combination of the important features into one system - Guardian Pass

A secure digital platform that combines digital gate-pass application, parent verification, warden approval, biometric authentication, QR-based entry/exit tracking and student safety support

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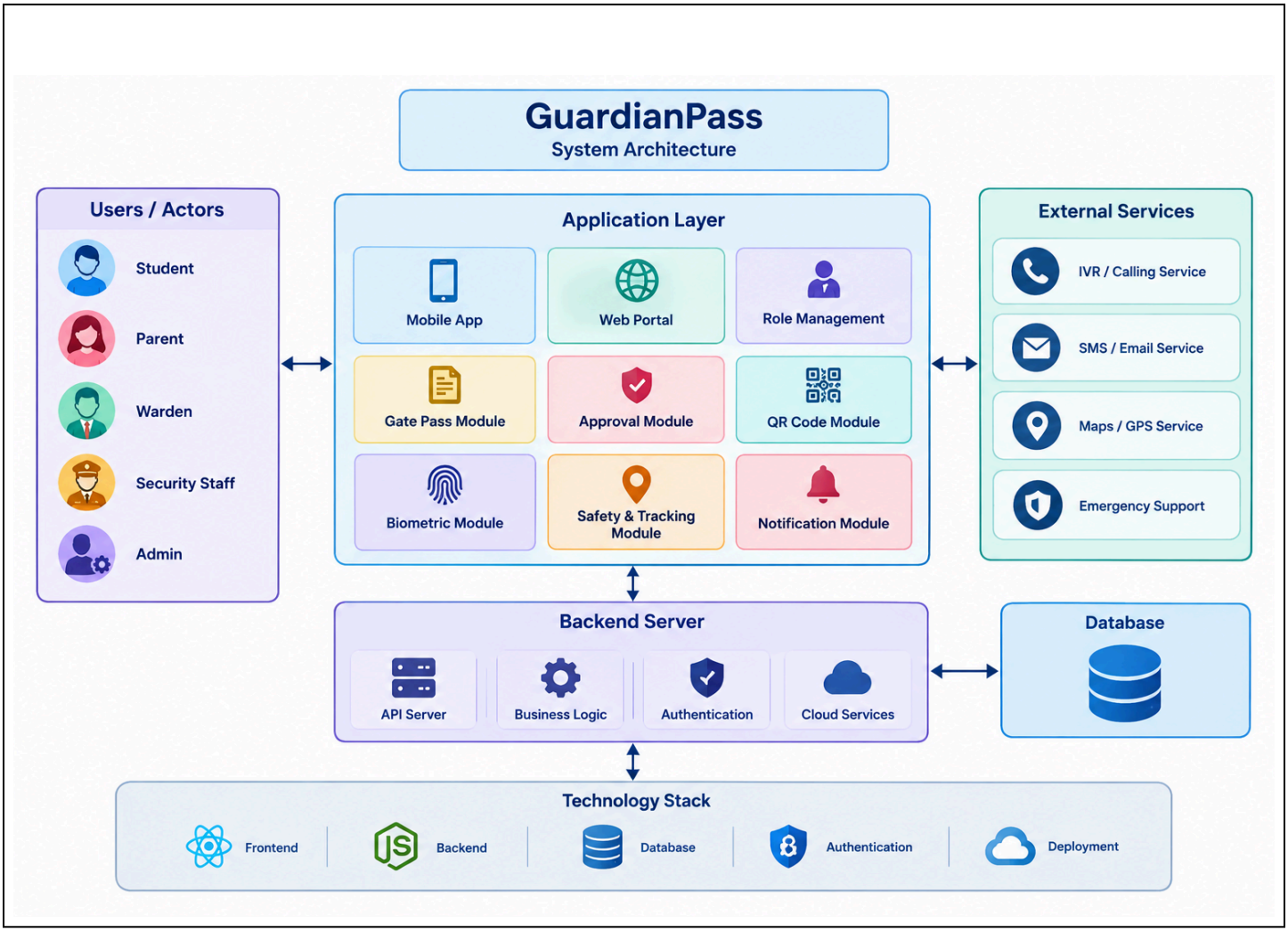
Concept Development

Feature	Description
Project Title	GuardianPass – Smart Gate Pass & Student Safety App
Key Features	**1. Digital Gate Pass** ● Online gate-pass application ● Pass status tracking ● Reason and destination details **2. Parent Verification** ● Parent approve/reject ● IVR verification option ● Approval timestamp **3. Warden Approval** ● Review student requests ● Approve/reject requests ● View active passes **4. Biometric Verification** ● Student identity verification ● Required before QR generation ● Verification result and timestamp **5. QR Gate Verification** ● Temporary QR generation ● Security QR scanning ● QR expiry and validation **6. Entry/Exit Tracking** ● Record exit time ● Record entry time ● Maintain movement history **7. Student Safety** ● SOS/Security button ● Emergency category selection ● Parent/authorized staff alerts **8. Parent Notifications** ● Approval updates ● Exit notification ● Entry notification

	• Late-return alerts **9. Location Awareness** • Location support during approved outings • Destination information • Safety monitoring **10. Late Return Monitoring** • Compare expected and actual return time • Notify relevant users when return time is exceeded
Technology Required	**Frontend** • React / Next.js • Tailwind CSS **Backend** • Node.js / Express **Database** • PostgreSQL / Supabase **Authentication** • Supabase Auth / JWT • Role-based authentication **Security** • Device biometric authentication • QR code generation and scanning **Location** • GPS / Maps API **Communication** • IVR / Automated Calling • SMS / Email / Push Notifications **Deployment** • Vercel • Supabase / Cloud backend
Expected Benefits	**For Students**

	• Faster gate-pass application • Less paperwork • Easy approval tracking • Secure identity verification • Quick emergency support **For Parents** • Direct control over permission • Timely movement notifications • Better awareness of student safety • Late-return alerts **For Wardens** • Easier approval management • Centralized gate-pass records • Reduced manual work • Better monitoring **For Security Staff** • Quick QR verification • Accurate entry/exit records • Reduced manual register work **For Hostel Administration** • Centralized records • Better transparency • Improved hostel safety management

Solution Architecture Diagram



Short Idea Pitch

Problem Addressed:

Hostel students face manual gate-pass applications, delayed approvals, difficult parent verification, inaccurate entry and exit records, limited safety monitoring, and delayed communication. Parents may not receive timely updates about student movement, late returns, or emergencies during approved outings.

Proposed Solution:

GuardianPass provides a digital gate-pass system where students apply online, parents and wardens approve requests, biometric verification confirms student identity, and a secure QR code manages entry and exit. Parents receive timely notifications, while safety features support tracking, late alerts, and emergencies.

Key Benefit to Users:

Hostel students may face delays and difficulties with manual gate-pass applications, parent verification, entry/exit recording and safety monitoring. Parents may also lack timely information about student movement and late returns.

Signature of the Coordinator